

MGT 634, ESAI, Homework #1 – Exploring Hospital Emergency Department Data

National Hospital Ambulatory Medical Care Survey (NHAMCS), Emergency Department, 2015

Overview

In this first homework assignment, you will work with a large, nationally representative dataset of emergency department (ED) visits in the United States. The data come from the 2015 National Hospital Ambulatory Medical Care Survey (NHAMCS). Your goals are to (i) successfully load and explore the dataset in Python, (ii) compute useful descriptive statistics, and (iii) interpret your findings through the lens of hospital management and operations.

You will turn in **three components**:

1. A **3–4 page written summary** (including tables/figures) that addresses the questions below.
 2. Up to **5 slides** summarizing your approach and key findings.
 3. Your **Python code or Jupyter notebook**, clearly organized and commented.
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Notes:

For any question requiring totals or rates, use the survey weight variable (PATWT).

For more information, visit: <https://www.nber.org/research/data/national-hospital-ambulatorymedical-care-survey>.

Assignment Questions (10 Parts)

1. Load and Inspect the Data

Load the NHAMCS-ED 2015 dataset.

- (a) Report the number of rows and columns.
 - (b) Identify variables with substantial missing data (e.g., using `df.isna().sum()`).
 - (c) What does each row represent?
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2. Weighted Totals and Sample Design

- (a) How many visits are included in the dataset (unweighted)? (*each row = one visit*)
 - (b) What is the estimated total number of U.S. ED visits in 2015 when applying the weight PATWT?. Confirm that your total aligns with the documentation (it's over 100 million visits).
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3. Basic Descriptive Statistics

Compute and interpret the following, both using unweighted visits as well as using weighted visits.

- a) Average and median patient age (*Column: AGE*).
- b) Fraction of visits by sex (*Column: SEX*) and by ethnicity (*Column: ETHIM*).
- c) Median waiting time to see a provider (*Column: WAITTIME*).
- d) Average length of visit (*Column: LOV*).

Discuss briefly what these statistics might imply about ED patient flow. Describe how weighting affects your interpretations of patient flow.

4. Demand Patterns by Time

Using the columns VDAYR (day of week) and ARRTIME (note the format):

- (a) Summarize the total (weighted) visits by day of week and by arrival hour. (b) Identify the busiest days and times.

What implications might these demand patterns have for hospital staffing or capacity planning?

5. Payer Mix

Using the column PAYTYPER (recoded primary expected source of payment):

- (a) Tabulate the weighted share of visits by payment type.
- (b) Compare the proportions of visits paid by Medicare, Medicaid, private insurance, and selfpay.

How might this distribution affect hospital financial performance?

6. Chronic Conditions

Using columns such as HTN, DIABTYP2, OBESITY, DEPRN, and TOTCHRON (look them up):

- (a) Compute the share of visits involving at least one chronic condition.
- (b) Compute the most prevalent chronic condition among patients (and the conditions you consider).

What does this suggest about the role of EDs in managing chronic disease?

7. Injuries and Intentionality

Using INJPOISAD (injury/overdose/adverse effect indicator) and INTENT15 (intentional vs. unintentional):

- (a) What fraction of visits were related to injury, overdose, or adverse medical effects? (b) Among those, what share were intentional versus unintentional?
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8. Diagnostic Services

Using column ANYIMAGE (any imaging) as well as columns such as CBC, GLUCOSE, BUNCREAT, ELECTROL, etc:

- (a) Among all (weighted) visits, what fraction included any imaging?
 - (b) Among visits with blood tests, which were most frequently ordered?
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9. Medications

Using columns such as NUMMED (number of medications) and MED1–MED30:

- (a) Compute the mean and median number of medications per visit (weighted).
- (b) Report the share of visits with no medications, 1–2 medications, and 3 or more

medications. What do these results imply about patient complexity and treatment intensity?

10. Correlations

Given your work above, compute what you think might be meaningful correlations among the variables you considered above as well as additional variables, including HDSTAT, DIEDED, DOA, etc. (look them up).